



VECTOR SPACES

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CLASS 5 : CONTENT



- Problems on Linear independence and dependence of vectors
- Problems on Basis and Dimension

LINEAR DEPENDENCE AND INDEPENDENCE

Problem 1: Decide the dependence or independence of the following vectors:

(1) vectors $(1, 3, 2)$, $(2, 1, 3)$, and $(3, 2, 1)$

Solution: Write the given set of vectors as columns of matrix A .

$$A = \begin{pmatrix} \textcircled{1} & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix} \xrightarrow[\substack{R_2 - 3R_1 \\ R_3 - 2R_1}]{\substack{R_2 - 3R_1 \\ R_3 - 2R_1}} \begin{pmatrix} \textcircled{1} & 2 & 3 \\ 0 & \textcircled{-5} & -7 \\ 0 & -1 & -5 \end{pmatrix} \xrightarrow{R_3 - \frac{1}{5}R_2} \begin{pmatrix} \textcircled{1} & 2 & 3 \\ 0 & \textcircled{-5} & -7 \\ 0 & 0 & \textcircled{-\frac{18}{5}} \end{pmatrix}$$

∴ All the columns of matrix A is with pivots

$S(A) = n = 3$ Given vectors are linearly independent.

LINEAR DEPENDENCE AND INDEPENDENCE

2) vectors $(1, -3, 2)$, $(2, 1, -3)$ and $(-3, 2, 1)$

$$A = \begin{pmatrix} \textcircled{1} & 2 & -3 \\ -3 & 1 & 2 \\ 2 & -3 & 1 \end{pmatrix} \xrightarrow[R_3 - 2R_1]{R_2 + 3R_1} \begin{pmatrix} 1 & 2 & -3 \\ 0 & 7 & -7 \\ 0 & -7 & 7 \end{pmatrix} \xrightarrow{R_3 + R_2} \begin{pmatrix} \textcircled{1} & 2 & -3 \\ 0 & \textcircled{7} & -7 \\ 0 & 0 & 0 \end{pmatrix}$$

∴ $\exists$ columns without pivots also in A.

Given vectors are Linearly Dependent.

3) vectors $(1, 1)$, $(2, 3)$, $(1, 2)$

$$A = \begin{pmatrix} \textcircled{1} & 2 & 1 \\ 1 & 3 & 2 \end{pmatrix} \xrightarrow{R_2 - R_1} \begin{pmatrix} \textcircled{1} & 2 & 1 \\ 0 & \textcircled{1} & 1 \end{pmatrix} \text{ vectors are linearly Dependent}$$

LINEAR DEPENDENCE AND INDEPENDENCE

Note: A set of 'n' vectors of $\mathbb{R}^m$ must be Linearly dependent if $n > m$.

4) vectors $(1\ 1\ 0\ 0)^T$, $(1\ 0\ 1\ 0)^T$, $(0\ 0\ 1\ 1)^T$, $(0\ 1\ 0\ 1)^T$

$$A = \begin{pmatrix} \textcircled{1} & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_2 - R_1} \begin{pmatrix} \textcircled{1} & 1 & 0 & 0 \\ 0 & \textcircled{-1} & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_3 + R_2} \begin{pmatrix} \textcircled{1} & 1 & 0 & 0 \\ 0 & \textcircled{-1} & 0 & 1 \\ 0 & 0 & \textcircled{1} & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

$\therefore$ All columns of A
are with pivots
Given set of vectors are
Linearly Independent.

$$\begin{pmatrix} \textcircled{1} & 1 & 0 & 0 \\ 0 & \textcircled{-1} & 0 & 1 \\ 0 & 0 & \textcircled{1} & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xleftarrow{R_4 - R_3}$$

LINEAR DEPENDENCE AND INDEPENDENCE

Problem 2: If w_1, w_2, w_3 are independent vectors, show that the differences $v_1 = w_2 - w_3$, $v_2 = w_1 - w_3$, $v_3 = w_1 - w_2$ are dependent.

Solution: $\because w_1, w_2, w_3$ are independent

$$\alpha w_1 + \beta w_2 + \gamma w_3 = 0 \Rightarrow \alpha = \beta = \gamma = 0 \text{ only}$$

To check if v_1, v_2 and v_3 are independent
check for the combination $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$

LINEAR DEPENDENCE AND INDEPENDENCE

$$\because C_1(W_2 - W_3) + C_2(W_1 - W_3) + C_3(W_1 - W_2) = 0$$

$$\Rightarrow C_1W_2 - C_1W_3 + C_2W_1 - C_2W_3 + C_3W_1 - C_3W_2 = 0$$

$$\Rightarrow W_1(C_2 + C_3) + W_2(C_1 - C_3) + W_3(C_3 - C_1) = 0$$

$$\Rightarrow \alpha W_1 + \beta W_2 + \gamma W_3 = 0$$

$\because W_1, W_2, W_3$ are linearly independent

$\alpha = \beta = \gamma = 0$ only

Hence dependent.

$$C_2 + C_3 = 0 \Rightarrow C_2 = -C_3$$

$$C_1 - C_3 = 0 \Rightarrow C_1 = C_3$$

$$C_3 - C_1 = 0 \Rightarrow C_1 = C_3$$

$\Rightarrow C_1 = C_3 = -C_2$
 C_i 's can assume values other than zero also.

LINEAR DEPENDENCE AND INDEPENDENCE

Problem 3: Find the Basis and hence find the Dimension of Subspaces of $\mathbb{R}^4$

1) All vectors whose components are equal.

∴ vectors are in $\mathbb{R}^4$ it should contain 4 components.

$$\begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} \in \mathbb{R}^4 \sim \begin{pmatrix} x \\ x \\ x \\ x \end{pmatrix} = \left\{ x \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}; x \in \mathbb{R} \right\} \text{ subspace}$$

Basis $(1 \ 1 \ 1 \ 1)^T$; Dimension = 1

BASIS

2) All vectors whose components add up to zero

$$\begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} \in \mathbb{R}^4 \text{ s.t. } x+y+z+t=0$$

$$\Rightarrow x = -y-z-t$$

$$\begin{pmatrix} -y-z-t \\ y \\ z \\ t \end{pmatrix} \sim \left\{ y \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + z \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} ; y, z, t \in \mathbb{R} \right\}$$

Subspace of $\mathbb{R}^4$

$$\text{Basis } \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\} \quad \text{Dimension} = 3$$

BASIS

Problem 4 :

Let 'V' be a subspace of four dimensional space $\mathbb{R}^4$

$$\text{So } V = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathbb{R}^4; x_1 - x_2 + x_3 - x_4 = 0 \right\}$$

Find the Basis and Dimension of V.

Solution :- $\because x_1 - x_2 + x_3 - x_4 = 0 \Rightarrow x_1 = x_2 - x_3 + x_4$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \sim \begin{pmatrix} x_2 - x_3 + x_4 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \sim x_2 \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$\underbrace{\hspace{1.5cm}}_{V_1} \quad \underbrace{\hspace{1.5cm}}_{V_2} \quad \underbrace{\hspace{1.5cm}}_{V_3}$

Any vector in subspace V of $\mathbb{R}^4$ can be obtained as a Linear combination of vectors (V_1, V_2, V_3) .

BASIS

Problem 5: Find a Basis for each of the following subspaces of 2×2 matrices.

- 1) All diagonal matrices
- 2) All symmetric matrices ($A^T = A$)
- 3) All skew symmetric matrices ($A^T = -A$)

Solution: Any 2×2 matrix is given by $\begin{pmatrix} a & b \\ c & d \end{pmatrix}; a, b, c, d \in \mathbb{R}$

Basis of 2×2 matrix is $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$

$\therefore \exists$ 4 vectors in Basis set

Dimension of 2×2 matrices subspace is 4

BASIS

1) All diagonal matrices (2 by 2)

$$\left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mid a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}; a, b \in \mathbb{R} \right\}$$

Basis is $\left\{ \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$ Dimension = 2

2) All symmetric matrices ($A^T = A$) (2 by 2)

$$\left\{ \begin{pmatrix} a & b \\ b & c \end{pmatrix} \mid a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}; a, b, c \in \mathbb{R} \right\}$$

Basis is $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$ Dimension = 3

BASIS

3) All skew symmetric matrices ($A^T = -A$) (2 by 2)

$$\left\{ \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix} / a \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}; a \in \mathbb{R} \right\}$$

$$\text{Basis} \left\{ \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\} \quad \text{Dimension} = 1$$

Problem 6: Find a basis for subspace of polynomials $p(x)$ of degree ≤ 3 .

$$\text{Solution: } \{1, x, x^2, x^3\} \text{ Basis, Dimension} = 4$$

SPAN OF VECTORS

Problem 7: Describe the subspace of $\mathbb{R}^3$ spanned by :

1) the two vectors $(1, 1, -1)$ and $(-1, -1, 1)$

Solution: Step 1: check the independence of given vectors before drawing conclusions w.r.t. subspace.

$$\therefore \begin{pmatrix} 1 & -1 \\ 1 & -1 \\ -1 & 1 \end{pmatrix} \xrightarrow[R_3 + R_1]{R_2 - R_1} \begin{pmatrix} 1 & -1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$\therefore$ Only 1 column with pivot, these vectors are dependent.
Hence, the span of the given vectors is a Line.

2) $(0, 1, 1)$, $(1, 1, 0)$ and $(0, 0, 0)$

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 \sim R_3} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 - R_1} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 0 \end{pmatrix} \xrightarrow{R_3 + R_2} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

SPAN OF VECTORS

- ∴ Only 2 columns with pivots,
the given set of three vectors span a
2 dimensional plane in $\mathbb{R}^3$
- 3) The columns of a 3 by 5 Echelon form matrix
with two pivots
2 Dimensional plane in $\mathbb{R}^3$
- 4) all vectors with positive components
$$\left\{ \begin{pmatrix} a \\ b \\ c \end{pmatrix} \sim a, b, c \in \mathbb{R} \right\} \sim \left\{ a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + c \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}; a, b, c \in \mathbb{R} \right\}$$

Whole of $\mathbb{R}^3$ i.e. 3 dimensional space.



THANK YOU

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